

ST512-SP26-Homework-1

Jason Menjivar-Hernandez

Problem 1

$$n = 36$$

$$\bar{x} = 23.4$$

$$sd(s) = 7.2$$

$$CI: \bar{x} \pm t^* \times SE$$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{7.2}{\sqrt{36}} = 1.2$$

$$t^* = qt(1-(.04/2), 36) = 2.13$$

$$CI: 23.4 \pm 2.13 * 1.2$$

$$96\% CI: (20.84, 25.96)$$

Problem 2

$$H_o : \mu = 3.5$$

$$H_A : \mu < 3.5$$

Will be conducting a left-tailed test because we want to know if the drug delivers relief faster (i.e., less than 3.5 minutes).

$$n = 50$$

$$\bar{x} = 3.1$$

$$s = 1.5$$

$$\alpha = 0.05$$

$$SE = \frac{s}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = 0.212$$

$$t\text{-stat} = \frac{\bar{x} - \mu_o}{SE} = \frac{-0.4}{0.212} = -1.887$$

$$p\text{-value} = pt(t\text{-value}, df) = 0.033$$

$p\text{-value} < \alpha$, we reject H_o

Therefore, there is sufficient evidence at the 5% confidence interval to conclude that the newly developed pain reliever delivers perceptible pain relief more quickly than the standard pain reliever.

Problem 3

	Design 1	Design 2
n_i	11	6
$\bar{y}_{i\cdot}$	52	46
sd	12	10

a

point estimate: $\mu_1 \hat{=} \mu_2 = \bar{y}_{1\cdot} - \bar{y}_{2\cdot} = 52 - 46 = 6$

95% CI: $(\bar{y}_{1\cdot} - \bar{y}_{2\cdot}) \pm t^* \times SE$

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{12^2}{11} + \frac{10^2}{6}} = 5.455$$

$df = n_1 + n_2 - 2 = 15$; assuming equal variance

$$t^* = qt(0.975, 15) = 2.13$$

95% CI: $6 \pm 2.13 * 5.455$

95% CI: $(-5.62, 17.62)$